Thinking like a business: Reconfiguring relationships to sustain open data infrastructures

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Abstract

Sustaining open data infrastructures over time is a complex puzzle, involving dynamic funding models and relationships with customers, collaborators, and competitors. Despite their importance, these mechanisms are often hidden from view, limiting their applicability to other infrastructures. In this article, we examine how Dryad, a well-known open data infrastructure, has worked toward financial sustainability by reconfiguring relationships with other actors and by strategically implementing a new business model and process of assetization. We identify four types of relationship reconfigurations with customers, collaborators, and competitors critical to Dryad's financial evolution: *reinforcing*, *forging*, *positioning*, and *excluding*. We then analyze how Dryad's strategic efforts to develop a new fee structure have changed its interpretations of value(s), community, and governance, factors important in an infrastructure's longevity. We conclude by highlighting emerging tensions that provide insight for other open infrastructures working to become financially sustainable. As a whole, our analysis focuses not just on financial mechanisms for funding open data infrastructures (although those emerge) but on the relationships which enable them.

1 Introduction

The long-term stability of knowledge infrastructures is one of the most persistent problems in information research. Despite this, their underpinning financial mechanisms, while critical (Borgman & Wofford, 2021), are often seen to be taboo or only become visible in times of crisis (Imker, 2025). Even in normal circumstances, financing is complex and precarious (Skinner et al., 2025), forcing many to focus on surviving rather than thriving (Borgman et al., 2020). Financial sustainability is further complicated by the relationships of individual infrastructures to other actors in the scholarly system, who can be customers, collaborators, or competitors.

We argue that understanding these infrastructural relationships, as well as the role of business models and asset creation within them, is critical to financial sustainability. While taking a business-oriented approach can be a necessary financial strategy, it is one that affects other critical factors important in sustainability. Financial planning must therefore take into account how business models and asset creation will affect other aspects of an infrastructure, such as its value(s), community, and governance. To make this argument, we draw on a mixed-methods study exploring how a well-known open data infrastructure (ODI), Dryad, has worked towards financial sustainability over nearly twenty years. We ask: which types of infrastructural relationships help or hinder financial sustainability? How are implementing business models and creating assets related to other aspects of sustaining infrastructure? What can other open infrastructures learn from the tensions which emerge when "thinking like a business?"

While financial security is a core concern, sustaining ODIs, or making them available for the long-term, involves many complex, inter-related factors. Tensions related to governance (Mounier & Dumas Primbault, 2023); maintenance (Borgman et al., 2016); standards (Edwards, 2010), and communities and usage (Fenlon et al., 2023; Morselli et al., 2025) all impact longevity. Sustaining infrastructure includes balancing the "infrastructural work" of managing content and keeping the lights on and the "institutional work" of building relationships with other actors (Mayernik, 2023). It is a constant process of balancing tensions between dynamic factors, as well as reconfiguring different forms of relationships (Edwards et al., 2007).

Relationships are also important in how business models, flexible collections of activities designed to create value (Amit & Zott, 2001), are implemented. Business models exert their own agency within an organization, shaping how practices and relationships between investors and consumers are coordinated (Fjeldstad & Snow, 2018; Zott & Amit, 2010). As business models are put into practice, organizations engage in processes of "assetization" (turning things into assets) (Birch, 2017) that involve a specific way of thinking, where objects or services are viewed as potential financial investments. This perspective, which we term "thinking like a business," in turn influences how assets are understood and governed (Birch, 2024).

We bring both of these perspectives – reconfiguring relationships and business thinking – to our analysis. After situating our work in relevant literature, we describe Dryad and how it has reconfigured infrastructural relationships to co-shape revenue streams over time. We identify four types of relationships with customers, collaborators, and competitors that have been critical to Dryad's financial evolution: reinforcing, forging, positioning, and excluding relationships. We then analyze how Dryad's strategic efforts to "think like a business" have shaped interpretations of value(s), community, and governance. We conclude by highlighting emerging tensions that provide insight for other open infrastructures working to become financially sustainable. Our analysis adds to the existing literature by focusing not just on financial mechanisms (although those emerge) but on the relationships which enable them and their effects on other aspects of sustainability.

2 Background

Data infrastructures can be seen as a form of knowledge infrastructure: complex assemblages of people, technologies, content, and policies (Edwards, 2010; Gregory et al., 2026). They are often "open infrastructures," characterised by the open sharing and reuse of data (Thaney, 2025), non-profit and community governance structures (UNESCO, 2021), or values related to equity and inclusion (Okune, 2019). While ODIs may seem stable, they continually reconfigure relationships with internal and external actors (Borgman et al., 2016; Karasti & Blomberg, 2018). Previous research has examined sustainability through the lens of relationships between infrastructures and the communities who use or produce them (Fenlon et al., 2023; Morselli et al., 2025); curation workers p(Thomer & Rayburn, 2024); and changes in technologies (Ribes et al., 2009). Relationships with funders and policymakers created through "institutional work" are also important in helping ODIs to transition from project status to becoming organizations (Mayernik, 2023). These relationships are manifest in numerous dynamic factors influencing longevity, which are at times framed as "risks" in digital preservation. We therefore situate our study of Dryad's financial sustainability in literature examining organizational or economic risk, as well as literature about business models, assetization and funding ODIs.

2.1 Risks and frameworks for sustaining ODIs

ODIs face numerous risk sources that are enumerated in digital preservation literature. Recker et al. (2026) classify risk factors as legal, technical, procedural, or organizational (related to economic failure). Organizational factors have been shown to result in the closure of data repositories. An analysis using the re3data registry, e.g.,identified nearly 200 repositories that had shut down (Strecker et al., 2023). Although the reason for most of these closures was unclear, organizational failures, i.e. the ending of project-related funding, were often a contributing factor. This supports earlier work focusing on life science repositories, where an even greater percentage of analyzed repositories were no longer available (Attwood et al., 2015). Notably, risk and closure are contextual and socially constructed (Frank, 2024), and shutdown can in fact be a positive, planned-for process of "convivial decay" (Cohn, 2016).

Although organizational and economic risks pose a threat, financial models and governance structures remain under-studied (Eschenfelder et al., 2022; Mounier & Dumas Primbault, 2023). Two recent grass-roots frameworks developed to help infrastructures mitigate these risks have received particular attention. The Principles of Open Scholarly Infrastructure (POSI) (POSI Adopters, 2025) provide guidelines for OIs to design governance, financial, and insurance strategies in line with the values of open infrastructure. POSI emphasizes that OIs should be transparent about finances, use short-term funds for short-term costs and use revenue sources in-line with an organization's mission. The Global Open Research Commons (GORC) (Jones et al., 2023; Treloar & Woodford, 2025) stems from work within the Research Data Alliance to create an interoperable "commons" of infrastructures, linked to each other through social and technical elements. GORC emphasizes the need for long-term financial models and governance structures that can change as KIs evolve. GORC in particular highlights the importance of maintaining interoperable connections and relationships with other infrastructures in the scholarly system.

2.2 Funding open data infrastructure

Relationships with consumers, stakeholders, and competitors also influence how open data infrastructures (ODIs) are funded. These relationships are managed and negotiated through the process of creating value and conducting business activities with different types of organizations (Birch, 2024), which have varying logics, business models, and processes of assetization.

2.2.1 Business logics, models and assetization

Proprietary companies and non-profit organizations are often portrayed in opposition to each other in scholarly communication, implicitly invoking classic views of organizations in management literature. In such views, commercial, shareholder-oriented firms operate using market logics to achieve financial return, having values of efficiency and competition (Friedman, 1970; Porter, 1985). In contrast, non-profit organizations are seen to foreground community, moral, or professional logics, espousing service-oriented values (Hansmann, 1980; Salamon, 1995). Importantly, more recent scholarship highlights that organizations are rarely purely profit- or mission-oriented, but are hybrid in nature, where different logics exist at the same time (Battilana & Lee, 2014; Skelcher & Smith, 2015). Typical tensions tend to emerge within organizations: in the case of non-profit firms, such as Dryad, tensions can surface between competing needs of achieving public missions and procuring funding or between service values and management activities (Anheier, 2014; Ebrahim, 2019).

Business models, important in all types of organizations, can be seen as flexible combinations of activities designed to create and capture value for particular assets (Amit & Zott, 2001; Zott & Amit, 2010). Such activities include developing value propositions, identifying costs, and experimenting with revenue streams. Revenue models are just one aspect of this complex combination of practices. Business models exert their own agency, shaping how organizations structure and coordinate work and relationships with actors, such as investors and consumers (Fjeldstad & Snow, 2018; Zott & Amit, 2010). Doganova and Muniesa (2015) propose that the flexible nature of business models allows them to act as boundary objects (Star & Griesemer, 1989), where they usefully circulate and are given meaning in local contexts. Within these contexts, business models are "performed" through pitches, tinkering with potential revenue sources, and serving as examples for other organizations (Doganova & Muniesa, 2015).

This constructed, process-centered nature of business models is reflected in critical scholarship regarding goods and assets. Here, goods are not fixed objects, but are rather made through processes of "qualification and re-qualification", where consumers and producers co-determine what is valuable (Callon et al., 2002). This process is particularly visible when selling services, e.g. in information economies. The distinction between goods and services is not always as important as differentiating assets, which generate durable rents or income, and commodities, which have a one-time price (Birch, 2020; Dreyfuss & Frankel, 2015). Turning something into an asset, a process Birch (2017) terms "assetization," involves reconfiguring relationships with different actors, claims and practices; this process changes how assets are understood, managed, valued and governed (Birch, 2024). Assetization therefore involves a particular way of thinking, of seeing things as (potential) financial investments and then relating to or "governing" them through this perspective (Birch, 2024; Muniesa & Doganova, 2020).

2.2.2 Business and revenue models for open data infrastructure

A foundational report from the Organisation for Economic Co-operation and Development (OECD) (2017) highlights the types of activities that should be involved in business models for ODIs. These activities center around clearly articulating value propositions, potential stakeholders, costs of data sharing and curation, and revenue sources. Eschenfelder et al. (2022) underscore the importance of flexibility in ODI business models, identifying two forms: Type A flexibilities involve large-scale dynamics, i.e. developing new products and services, whereas Type B flexibilities, often more important, involve minute adjustments to revenue streams.

ODIs are often funded through a mix of revenue streams, e.g. combinations of institutional and project funding (Kitchin et al., 2015; Organisation for Economic Co-operation and Development (OECD), 2017), with public funding being consistently important (Attwood et al., 2015; Imker, 2025). Revenue sources for ODIs should ideally be in line with open principles and values (POSI Adopters, 2025; Shankar & Eschenfelder, 2015). The OECD report advocates that charging data depositors and charging for services (rather than access) best supports open values (Organisation for Economic Co-operation and Development (OECD), 2017). Despite this advice, selling services is relatively rare for data repositories (Eschenfelder et al., 2022). Other pragmatic factors such as timelines, eligibility, and legal restrictions also shape the ideal mix of ODIs' revenue streams (Hooft & Roos, 2025). Although revenue streams should cover the costs of an ODI, the costs of data sharing are often unknown or not delineated (Mohr et al., 2024). Costs can include a mixture of maintenance costs (e.g. computational resources) and soft costs (e.g. data curation), which are more difficult to determine (Borgman & Groth, 2025) but which can be significant (Uffen & Kinkel, 2019).

Overall, the literature indicates that sustaining ODIs involves relationship work: managing relations with other actors and systems, as well relations between various risk factors. Funding ODIs is a similarly dynamic process. Business models are flexible systems of activities; in the context of ODIs, this includes searching for revenue streams matching open values; determining costs for data sharing; and engaging in processes of assetization, where data or services are turned into assets capable of generating recurring value (Birch, 2017). We bring both of these perspectives – the importance of relationships in sustaining infrastructure and business thinking – to explore how Dryad has worked towards financial sustainability over nearly twenty years.

3 Methods

To explore Dryad's (financial) evolution, we designed a mixed methods study combining semi-structured interviews, document and material review, and a quantitative descriptive analysis of Dryad's data and data references.

3.1 Interviews and document review

We conducted 14 one-hour, online semistructured interviews between April and August 2025 with current and past Dryad staff and members of the Board of Directors. Interviews lasted approximately 60 minutes and were transcribed for analysis. To recruit participants, a relationship was first developed with Dryad senior staff, who agreed to support the study. Participants were then recruited based on recommendations and snowball sampling. Recruitment stopped once we had spoken with representatives from all professional roles at Dryad who were willing to participate in this study. In sum, we spoke with two full-time curators and technologists, eight board members (split equally between publishing and institutional representatives), and three (former) staff. We also spoke with one employee at another OI with extensive experience of Dryad. Speaking with these individuals enabled us to collect enough data to answer our research questions and achieve sufficiency in our data (LaDonna et al., 2021; Vasileiou et al., 2018). We describe demographics at an aggregate level to protect the anonymity of participants, and refer to participants by number (PX) in the findings.

Participants were invited to bring a document to serve as a discussion prompt; these included blogposts, reports, and policy documents. Participants explained the importance of these prompts and their relationship to sustainability. Other interview questions addressed different factors influencing the sustainability of ODIs, identified from the literature (Section 2.1), e.g. funding, governance, communities, and relationships to other repositories. Questions were tailored to match participants' expertise. ATLAS.ti ("ATLAS.ti", 2022) was used to support a combination of deductive and inductive coding and reflexive thematic analysis (Braun & Clarke, 2006, 2019).

We developed the codebook iteratively in rounds of discussion with both authors. After initial deductive code development, both authors coded a sample of three transcripts; codes were then further discussed. The authors then re-coded the set of transcripts and had a further discussion. The first author then coded the remaining transcripts; the second author read the remaining transcripts to feed into further discussion and iteration. Deductive codes were based on literature about sustaining knowledge and data infrastructures (Section 2.1) as well as the interview protocol. Deductive codes related to governance (e.g. community governance; board organization); funding (e.g. descriptions of the new fee structure); Dryad's relation to other repositories; and community needs and engagement. Inductive codes included, e.g., reconfigurations of particular relationships and business thinking. These inductive codes informed the framing of our analysis. The interview protocol and codebook are openly available, along with our quantitative data [reference removed for review]. This study is part of a project that has received ethical approval from [removed for review] University.

We also closely read openly available documentation about Dryad. This material included blogposts and news items written by Dryad staff, organizational bylaws from 2016 and 2024, a selection of minutes from board meetings and annual reports, and information on Dryad's website and related websites. We also attended two open webinars in spring 2025 for publishers and institutions explaining the new fee structure. This material was not systematically coded, but informed conversations with interviewees and nuanced our understanding of Dryad's development over time.

3.2 Descriptive data analysis

We conducted a quantitative analysis of Dryad's data to characterize the repository and understand the use of Dryad's data. This supported the qualitative methods. We analyzed Dryad's metadata, including subject fields and data references. We further analyzed Dryad's federal funding over time using an openly available dataset (Riordan et al., 2025); we detail these steps below.

DOIs were retrieved from the Dryad API on 2025-06-24[1]. At that time, 64.774 DOIs were accessible through the API. Metadata of datasets (author last names, publication year, publisher name and resource type) were added via the DataCite API on 2025-07-11[2]. The resulting sample includes metadata for 64.774 Dryad datasets. There are several sources that capture citations of datasets, relying on different strategies to identify data citations. To ensure that publications referencing Dryad data were covered as completely as possible, three sources were combined (Table 1). These sources were searched for the 64.774 DOIs retrieved from the Dryad API in the previous step.

source	date	method	number of citations
Data Citation Corpus v3.0 (DataCite & Make Data Count, 2025)	2025-02-01	Dryad DOI strings were matched with the "dataset" column	46.873
DataCite REST API - relationships section[3]	2025-07-11	“citations” were retrieved from the relationships section	54.335
OpenAlex[4]	2025-07-04	Dryad DOIs were passed as an argument to the "cites" filter	492

Table 1

Overview of sources used to retrieve references to Dryad datasets.

References were deduplicated. DOIs of all publications referencing Dryad data were checked; 318 were removed because the DOI did not resolve. This resulted in 50.126 references in total. Similar to previous studies, the type of relationship between the Dryad dataset and the referencing publication was determined by comparing author last names (Cohen et al., 2026). Following Bobrov et al. (2026), if there was overlap in author lists, the relationship was categorized as “data use" and “data reuse” if there was not. The comparison was based on author last names, due to limited availability of author identifiers. As last names may not uniquely identify authors, it is possible that this approach led to underestimating data reuse. Because the publisher field allows free text, the values are very heterogenous. Therefore, publisher names of referencing publications were harmonized and checked. Mentions of metadata aggregators and missing values were corrected. Publisher imprints were then grouped according to Haupka et al. (2025). This allowed us to group referencing publications at the company level, matching descriptions of publishing partners in Dryad's documentation.

For subject information, only labels assigned from the OECD Fields of Science Classification (FOS) were considered, as they were the most widely used controlled vocabulary in the sample. These labels were aggregated at the highest level of the classification. Each label was counted once per dataset. In total, 24.305 FOS labels of 24.148 datasets were included. To analyze Dryad's federal funding, data from the 2025 State of Open Infrastructure Report (Riordan et al., 2025) was reused. Eight direct grants awarded to Dryad with information about start and end date were included. The total amount for each of these awards was divided by the number of project months.

4 Findings

We begin by describing Dryad and the evolution of its financial model by tracing changes in key relationships with other actors. We then explore how the strategic development of Dryad's new business model and process of assetization evoked changes in various elements important in sustainability: namely, interpretations of value(s), notions of community, and governance.

4.1 Reconfiguring relationships for sustainability

Dryad is an "open data platform and a community committed to the open availability and routine re-use of all research data" (Dryad, n.d.-b). Originally rooted in the life sciences, Dryad portrays itself as a generalist data repository that accepts data regardless of type, format, or disciplinary focus (U.S. National Library of Medicine, n.d.). Despite this, the vast majority of datasets in our analysis continue to be related to the life sciences (Figure 1 a), with 74.5 % of datasets in the largest subject category, natural sciences, coming from the biological sciences. This category continues to see the highest number of deposits each year. One participant emphasized that being a generalist repository is not just related to content, but also to the type of (more general) metadata and curation activities provided. Participants mentioned that becoming a generalist repository was a business decision that allowed Dryad to broaden its potential customer base.

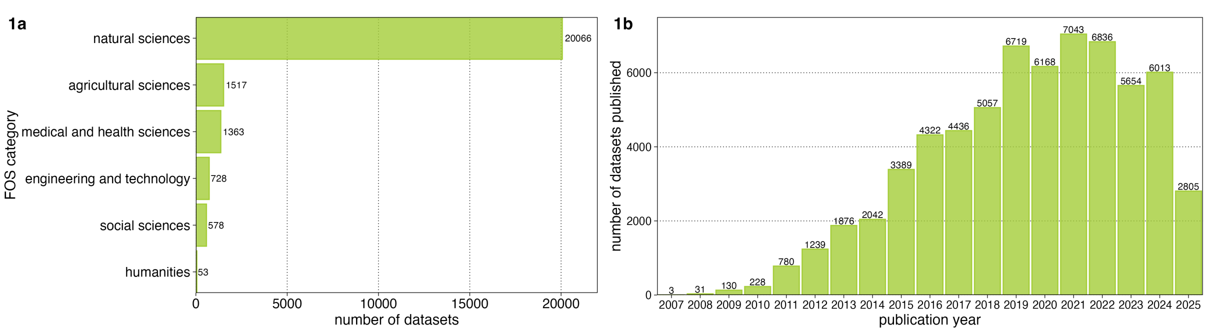


Figure 1

Characterization of datasets in Dryad. Figure 1a. Subject classification of datasets in sample according to FOS categories. Figure 1b. Number of datasets in Dryad over time.

Figure 1 b demonstrates an overall growth in the number of deposited datasets. There have been moments where growth plateaued, reflecting a slower pace of research during COVID-19, as well as key moments in Dryad's history. Many of these key moments were related to reconfigurations in relationships and revenue streams (Figure 2). Figure 2 demonstrates a common pattern in the development of ODIs: transitioning from a time-limited project to institutional embedding to becoming an independent organization (Organisation for Economic Co-operation and Development (OECD), 2017). We refer to Figure 2 throughout this section as we detail these reconfigurations.

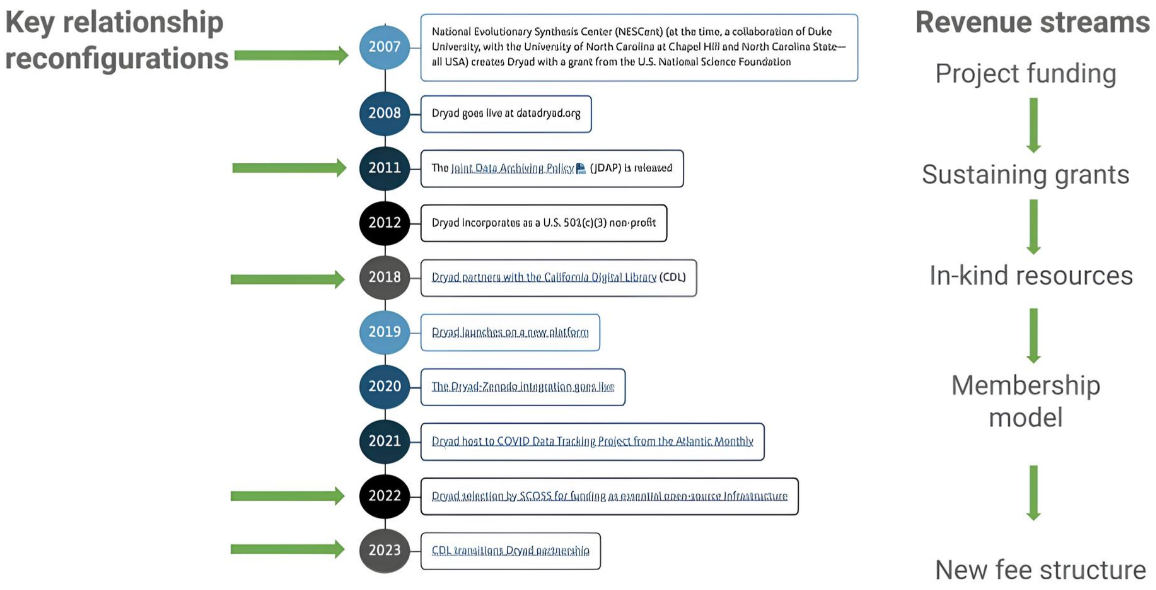


Figure 2

Key moments of relationship and revenue reconfigurations over time. The timeline was created from a screenshot on Dryad's webpage (December, 2025). Annotations to the timeline were added by the authors.

Dryad has its origins in a group of researchers at the National Evolutionary Synthesis Center, which was funded by the United States National Science Foundation (NSF). These researchers proactively created the first version of Dryad to meet publishers' and funders' data sharing policies. Initial project funding came from the researchers' institutions as well as dedicated grants from the NSF. The grassroots nature of these beginnings resonated with the broader OI community:

> [There were] other people who are in the story, working at other places who saw Dryad as this exemplar of the community rising up to meet the challenges of the day. You know, data policies [..] were all taking hold and people were like, "How do you address it?" And Dryad, as an example of researchers, saying, "Let's just build something ourselves for ourselves," was always a cool idea. (P7)

The development of Dryad reflects the complexity of funding streams typifying OI. This complexity is also visible in federal funding over time. Project funding was an important revenue source throughout Dryad's history, with a particular reliance on NSF grants (Figure 3 a). Although consistently present, the amount of monies from federal grants has varied (Figure 3 b), with a notable drop corresponding roughly to Dryad's relationship with the California Digital Library (Section 4.1.2).

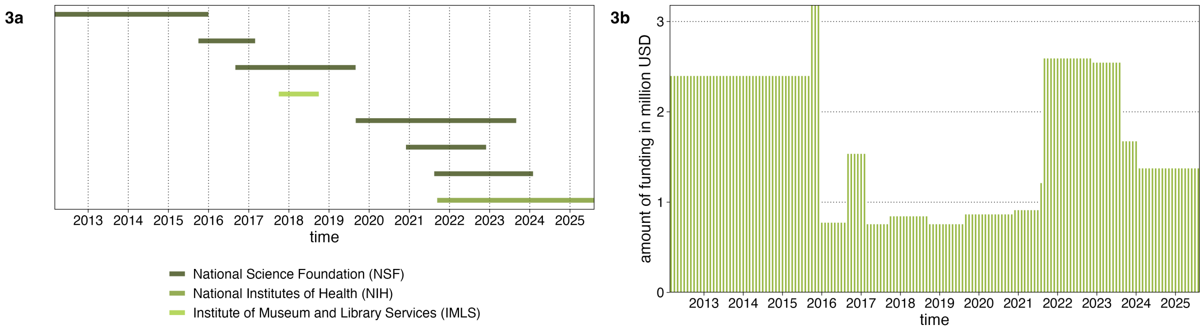


Figure 3

Federal funding for Dryad over time. Figure 3a. Funding streams according to funding agency. Figure 3b. Combined funding amounts across funders

Our interview data demonstrate different ways in which Dyrad has reconfigured relationships with other actors, both customers as well as (potential) collaborators and competitors, as it has worked to sustain itself. We classify these as reinforcing; forging; positioning; and excluding relationships. By nature dynamic and overlapping, this classification provides insight into the relationships supporting ODIs over time.

4.1.1 Reinforcing relationships with customers (publishers)

Strong relationships with publishers have been key to Dryad's persistence. Seeing the need to make data supporting publications available, but also lacking a viable way to do so, publishers encouraged the development and use of Dryad. In 2011, e.g., the Joint Data Archiving Policy (JDAP) was adopted in a coordinated fashion by many leading journals to require data sharing for published articles (Dryad, n.d.-a). Many journals recommended the use of Dryad in their statements. Notably, the policy was not released until Dryad was operational. As one participant described:

> There were two things that happened; the centre obtained additional funding from the National Science Foundation specifically to build Dryad. And then there was a coalition of these journals that said: as soon as Dryad is ready to go, the journals will start requiring that all of our articles have associated data published. And the journals simultaneously launched this policy [..] because they didn't want any one journal to suddenly launch this new strict policy, and then have all the authors flee to the other journals. (P5)

This relationship with publishers became more seamless with time, as publishers embedded Dryad in their workflows. Technical integrations with publishing systems, such as automatic metadata completion for deposited data, have strengthened this relationship. This embedding is also visible in the citation of Dryad datasets in published articles; 69.18 % of Dryad datasets have been cited at least once. The vast majority (93.6 %) of data citations classify as instances of "use", meaning that the datasets were cited by data creators. This pattern holds across the top publishers in our sample (Figure 4), which includes five publishers listed as "founding partners" on the Dryad website: Wiley, the Royal Society, Oxford University Press, PLOS, and AAS, demonstrating the longevity of these relationships.

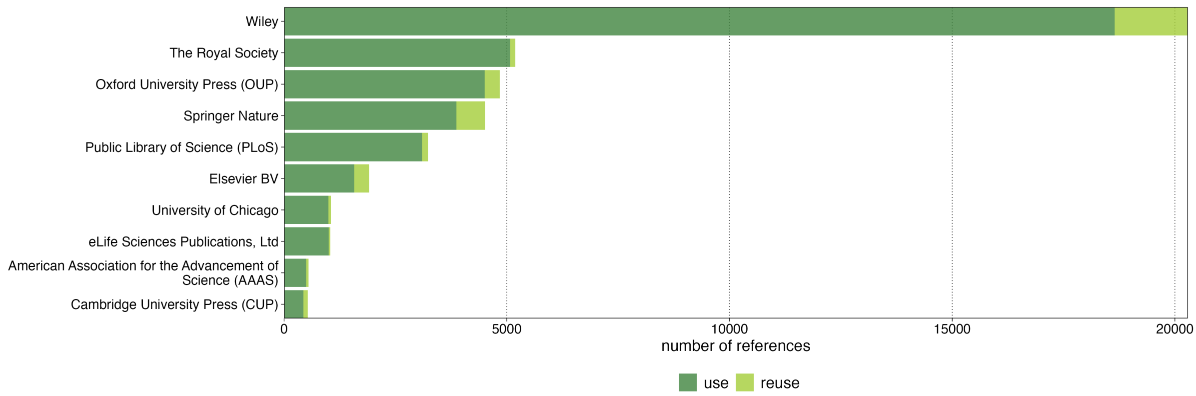


Figure 4

Number of references classified as use and reuse for the top ten publishers in our sample. Use indicates that the datasets and publications were classified as being created by the same author. Reuse indicates that a dataset and publication were created by different individuals.

4.1.2 Forging relationships with customers (institutions)

In addition to reinforcing relationships, Dryad has forged new relationships, most notably with institutions. Some of these relationships were carefully cultivated to reach new institutional customers; some were more serendipitous, as in the case of a core relationship in Dryad's history: its partnership with the California Digital Library (CDL). This partnership began at a moment when Dryad was struggling to make ends meet, in part because of technical limitations. At the same time, CDL was developing an institutional repository that felt like a siloed and under-used resource. Informal, honest conversations about the struggles of funding open infrastructure created an environment for "radical collaboration":

> Open infrastructures succeed when you—without any ego or pretence, just can open your doors to the opportunities that come with openness. So, really thinking not just about planning for openness, but actually just being there for serendipitous or opportunistic openness and collaboration. (P8)

This new collaboration was key for Dryad's survival, as suggested by the increase in deposited datasets at this time (Figure 1b). CDL provided in-kind contributions in the form of dedicated staff for technical development, product management, outreach, and strategy. These in-kind contributions were seen as critical for moving Dryad to a new technical platform and expanding the existing business model.

> We weren't just talking about technology. We were also talking about business models. From the very beginning, we were saying, "For this to work, Dryad needs to change. Not just its product and its platform, but also its offering." [..] Before they did not publish datasets, unless they were attached to a journal article. And we were like, from an institutional perspective, we have datasets that don't have that [..] We would need features—administrative features—that would help with institutional use cases. And that kind of a thing meant that they had to agree to change from a business model before CDL even signed. (P8)

The relationship with CDL, based on a shared sense of values, widened Dryad's reach to enroll other institutional members.

> Institutions were key to Dryad's success because institutions shared values with what Dryad was presenting. When CDL convinced everybody to take this path and make this leap and say, "Why don't we just work together and [..] all ships will rise," the thing that made it happen was saying, "We share values. Here are our institutional values and here are Dryad's values. They are the same." I think that that's also a piece of this puzzle [..] libraries and institutions and open infrastructure, they share the same values. (P7)

4.1.3 Positioning relationships with collaborators and competitors

Dryad has also engaged in positioning relationships, where it positions itself alongside or differentiates itself from certain infrastructures, particularly data repositories, scholarly communication infrastructures, and open/proprietary infrastructures.

Positioning with data repositories
Dryad's positioning varies in relation to different types of data repositories, be those disciplinary, institutional, or other generalist repositories. Numerous participants reiterated the importance of having repositories with specific metadata and curation workflows for data from particular disciplines. Dryad positions itself as complementing these disciplinary repositories, as a place for heterogeneous data. In contrast, Dryad clearly positions itself as a replacement for institutional repositories. Providing curatorial and storage services for institutions is seen as a resource-efficient way to help institutions meet the needs of their researchers (Section 4.2.1). At the same time, other generalist repositories, such as Zenodo or Dataverse, are seen as "peers".

> I think that many people [..] would say, "Do we really need Dryad and Zenodo if they're both open or something like that?" When in reality, it's not about that. What the question in that case is, why do we have all these data institutional repositories when we have things like Dryad and Zenodo? [..] Why are there thousands of [institutional] data repositories around the world with ten datasets in them? (P7)

Positioning with other scholarly communication infrastructures
Dryad has engaged in the institutional, project, and policy work necessary to embed itself within

the broader scholarly ecosystem. In 2022, Dryad became part of the Global Sustainability Coalition for Open Science Services's (SCOSS) list of essential infrastructures, which came with extra funding and facilitated connections with other SCOSS "family members" (Figure 1). This map depicts technical integrations and joint projects between Dryad and generalist data repositories, societies, persistent identifier providers, and publishing organizations. These embeddings anchor Dryad in a web of related actors across the scholarly system.

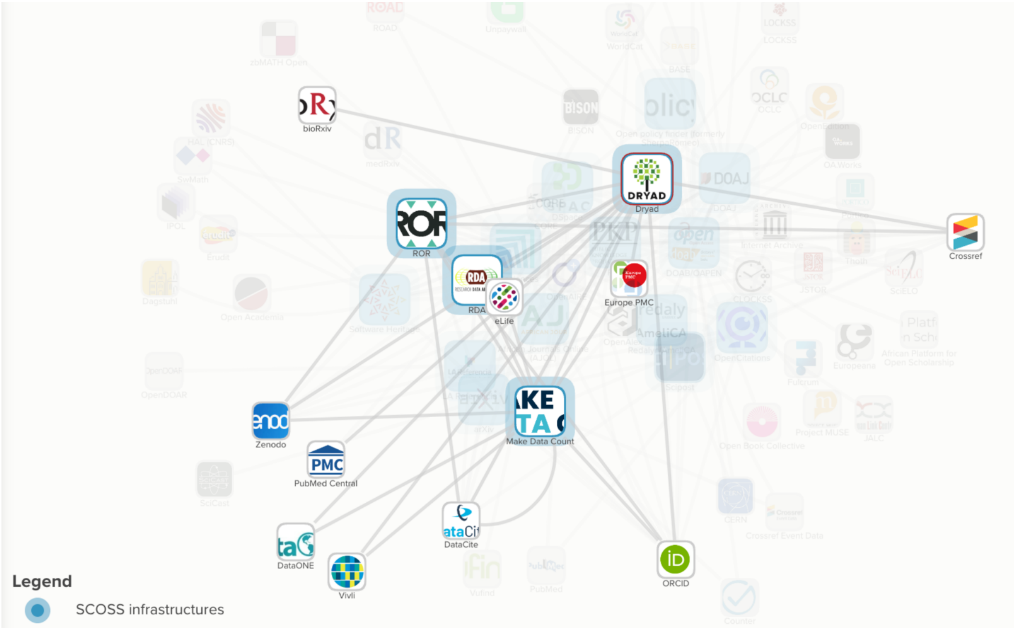


Figure 5

Technical and project integrations between Dryad and other SCOSS "family members". Map retrieved February 11, 2026 from https://scoss.org/what-is-scoss/scossfamily/

Positioning with open/proprietary infrastructures

Dryad carefully positions itself as an open infrastructure, often in opposition to proprietary or commercial infrastructures. Openness is framed in different ways, ranging from publishing all data openly using a CC0-licence to being transparent about governance documents, to having a "nonprofit ethos".

> I think of all of the generalist repositories, Dryad is the one that's most attractive personally, because they do have the non-profit ethos of "we're doing this to support good science and advance research," not to make a buck. And certainly, they need to make a buck and they want to make a buck. But that's to sustain the open infrastructure and the services not to go retire on the Riviera or whatever else everyone does. (P2)

Openness of technology is also seen as a differentiating factor between Dryad and both proprietary infrastructures as well as other nonprofit organizations.

> So, it's not only that we're nonprofit [..], it's the operation of the technology.

> My understanding is that open infrastructures are not necessarily run by open organisations. Hopefully they are, because the values are consistent in that way, but open infrastructure is where the software is open itself [..] And that stands in contrast to a commercial organisation or another nonprofit organisation that might keep its code closed. (P1)

The previous quote suggests that the line between open and non-open is not always clear cut, but exists instead on a spectrum. It also highlights the importance of "open values" (Section 4.2.1) in how Dryad positions itself; many participants see these values as a necessary differentiating factor which enables Dryad to compete against proprietary infrastructures with more resources.

> It's sort of in competition with private publishers, where you have some private organisations that are quite big [..] And they can operate in this space because they have a very different fiscal model [..] And there's a danger of private interests sort of out-competing public repositories .. [but] a lot of people do value the fact that we're not a private company that has private interests to just provide a product, but we actually really care about the values of openness. (P10)

In its positioning relationships, Dryad faces questions about competition and collaboration. Dryad positions itself as a competitor to institutional repositories or in opposition to proprietary infrastructure. In other relationships, collaboration and partnership are seen as a strength, e.g. in thinking about how to maximize limited financial resources within the OI community.

> This is a conversation [..] about the multiplicity of open infrastructure, all knocking on the same funders' doors to get money. How can we all either share resources or come up with a package that says, "This is the support that open infrastructures can provide to you as an academic institution." And how do we collectively create those synergies so that people don't go and pick a proprietary software? (P6)

Sometimes, as in the General Repository Ecosystem Initiative (GREI), competing repositories collaborate "below the value line" in a co-opetition model. This promotes base level standards in the broader ecosystem without compromising (or revealing) the unique value-add of an individual repository.

> What our experience has been is folks keep close to the vest, whatever they keep close to the vest, but we are able to engineer productive collaboration around getting all of these platforms to adopt DataCite as the one metadata schema. That hadn't been done before [..] There's broad agreement that that stuff is not going to win business for one repository over another (P1)

4.1.4 Excluding relationships with collaborators and competitors

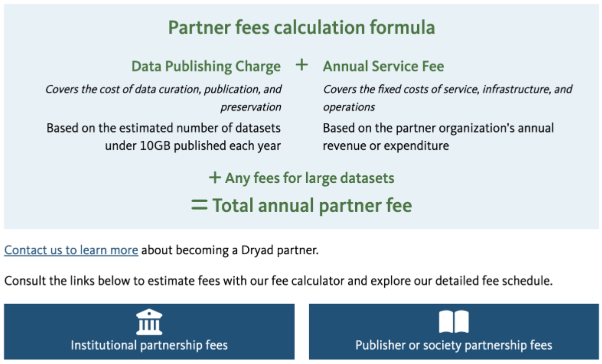
Dryad makes considered choices about the types of relationships to pursue but also about those to avoid or end. Participants described avoiding relationships with organizations who had been purchased by proprietary firms. Another example is the conclusion of Dryad's relationship with the California Digital Library. Frictions concerning differing visions of financial models contributed to the ending of this relationship. Despite these frictions, both organizations (who are prominent in the OI community) saw value in concluding the relationship in an amicable way.

> We took as an exec group the decision that we need to change the model to actually get to a model that would be sustainable. And so that meant actually having costs that were effectively metered by usage [..] Now, we did have a strong disagreement on the sort of philosophical direction of that with our partners at CDL. [...] We got to an amicable place where they said, "OK, we understand what you're doing. We would prefer you to try to continue to work in this way. But if you want to operate independently, then how do we make that work in a way that can take both organisations forward?" (P11)

4.2 Thinking like a business: Shaping sustainability

The previous section demonstrated how reconfigurations of relationships with customers, collaborators and competitors have contributed to Dryad's financial sustainability. We now explore how thinking like a business, by implementing a new business model and engaging in a process of assetization, has affected not only financial sustainability, but also other elements important in sustaining infrastructure: value(s), community and governance.

Developing Dryad's new fee structure (Figure6) grounded this process of business thinking. The new fee model consists of three primary parts: a data publishing charge to cover the variable costs of data curation and preservation; an annual service fee to cover fixed operational costs (e.g. training, administration, and leadership); and an extra fee for large datasets. These fees are paid by institutions or publishers/societies. The model differentiates between sponsored authors, from affiliated institutions or publishers, and unsponsored authors who pay individually or are granted a fee waiver. In cases where an author is affiliated with both a publisher and an institution, the institutional members are first asked for payment.



https://datadryad.org/join_us; Retrieved 05 March 2026

Dryad's new fee structure (rolled out in 2025) consists of two primary parts. The variable data publishing charge, based on the amount of data an institution or publisher deposits annually, covers variable costs such as data curation and preservation. The fixed annual service fee covers fixed costs, such as the cost of leadership and technology. The calculation of the service fee varies according to the size, revenue, and expenses of the customer's organization. This model was developed to charge for using Dryad (in terms of data deposits), cover actual costs, and separate fixed and variable costs. There is in additional component of the structure, not shown in the image, for contributions from open science supporters.

Figure 6

Description of Dryad's new fee structure. Image is a screenshot from Dryad's website in March 2026.

The process of developing this structure involved a three-year period of market research and consultation centering on articulating a value proposition. While taking a business-focused approach was seen to be necessary for Dryad's future, it was not something that came naturally to all actors:

> I think a unique thing that sort of happens in science [..] is that most scientists, including—I will still say myself, are not business people —like, not wanting to come in and look at numbers and books. (P10)

Dryad drew on the expertise of board members, including those with publishing backgrounds, a special financial task force composed of stakeholder representatives, and a financial consultant to re-evaluate Dryad's product market fit. It aimed to develop a fee structure meeting stakeholders' diverse interests that would provide a stable revenue source and support principles for open infrastructure.

4.2.1 Re-interpreting value(s)

This process involved interviews with stakeholders (customers), competitive analyses, literature reviews, and open consultations to understand and articulate the value that Dryad offered and could offer. Defining the product value of the repository was difficult; in part because of the immaturity of the "business case" for data sharing. Dryad's new value proposition ultimately coalesced around two key points: the idea of operating for "the community" and shifting from being a platform to a "service provider".

> Until we completed the value proposition work last year, the spiel was that Dryad is an open data publishing platform. [..] [Now] We want to be more clear that

> we are a nonprofit organisation operating in the community interest, that we are a community-owned organisation as well as being nonprofit. [..]. And the other thing we want to better articulate about our value proposition is that we are a service provider. We will operate in the community interest, but we are a trusted and reliable service provider that makes sure that data is prepared and shared and published and preserved appropriately. (P1)

The framing of this service varies for different stakeholders. For publishers, the service is sold as a way to easily support data sharing mandates. For institutions, the framing focuses on providing data curation. Charging for these services involved determining and attaching monetary value to these activities, effectively turning them into assets to generate longer term income. Even though many admitted that the curation Dryad provides is relatively lightweight, it was also seen as being a valuable service worth payment.

> Curation as an activity is very expensive [..] We at the libraries did not take that on, and so the amount of curation that Dryad provides – which I know is still minimal – is extremely attractive, because it means you don't have to have people on staff who are opening data sets and taking a look and reading and processing and those kinds of things. And even just the cursory sort of does the data match? That's really important [..] So having someone who includes in their publishing fee that service was really attractive. (P4)

Participants mentioned multiple "unique selling points" (USPs), above all curation and openness. Various tensions arise around these USPs. While curation is both an asset and USP, it is carried out by a very small staff, involving part-time employees located in different countries. Some participants mentioned the potential of AI tools to take over basic curatorial tasks and reduce labour costs. As one curator explained, it is not just the curation that is of value, but the conversations and connections with researchers enabled through curation activities. A key point of value is therefore human connection, in addition to curation.

The openness of Dryad's data and its source code were seen to be another USP. One participant discussed a tension between the perceived value of openness and convincing customers to pay for open infrastructure.

> A challenge that we've had at Dryad has been how institutions – picking again on the academic institutions case – very often see the value of organisations like Dryad and the service we provide. And they really appreciate that we're an open infrastructure [...] that we're community-led, and that we're not for profit [..] And in many cases, they appreciate the need for organisations like ours to be financially sustainable over time. But there is always a tension between perceived value and the money that people are willing to invest in open infrastructure. [..] I think "openness" can help with the philosophical cause for open infrastructure, but at the practical level, it's often a barrier. (P6)

While openness is seen as a USP and important value of Dryad, it also complicated the development of the fee structure. This is apparent in the above quotation and in decisions about how to generate income when code and content are openly available. This tension informed the development of the value proposition – where Dryad is a provider of services rather than content – and foregrounds the importance of human support and connection.

> Certainly it is more challenging for Dryad to have certain methods of funding because our code is open [..] So, we really centre a lot of our charges in our business model on the idea of our hands-on human staff that helps people through the process. And so everything we write in the software is in support of that, both to help the content be more well-structured before it even gets to our humans and for our humans to be able to deal with stuff after they receive it. (P5)

This section demonstrates the process of assetizing curation services and how the fee structure reflects certain values. Curation is associated with a value of care and human connection; and openness with transparency and accessibility. Notably, many discussions around these values were related to institutional stakeholders rather than publishers. This observation could reflect the already well-established relationships to publishers (in that there is no longer a need to use shared values as a selling point; Section 4.1.1) or the common practice of charging for services in publishing.

4.2.2 Evolving notions of community and governance

Operating in the community interest is part of Dryad's value proposition; involving stakeholders in the development of the fee structure was therefore crucial. Community involvement and transparency were leveraged to achieve support from (institutional) stakeholders, demonstrate the legitimacy of the pricing structure by making costs visible, and get further input into customer needs.

> We didn't go away in a room with just the Dryad team and run the numbers and figure out how much we need to cover our costs. [..] We convened a working group [..] that was mostly comprised of librarian members. We basically opened the books to them and said, look, this is how the organisation is structured. Here are the infrastructure costs. You know, curation is a key value. That's what curation costs. As representatives of the community, what do you want? (P11)

The composition of these meetings indicates who is seen to be the primary "community" for Dryad. While researchers were originally core members, the community has evolved to focus on stakeholders or customers: actors who will pay for Dryad's services, including institutions, publishers, and professional societies. To some degree, these actors are proxies for researchers, representing researchers' needs in addition to their own interests. The evolving composition of Dryad's community is also reflected in governance structures. Dryad's board includes members with institutional or library expertise and those with publishing backgrounds. Community governance in this case is not about "chasing you down to vote on someone you have never met" (P1), but rather having a selected expert group to steer the organization.

> It does make sense when you have a service to have a user group that gives you feedback: is the service evolving in the right direction for it to be useful for the intended purpose? [..] We have this board of directors, which I think historically was a bit of community governance. And now it's kind of developed along the business concept to get people in with different expertise who can help guide and develop Dryad further. (P14)

The above quotation demonstrates how Dryad's governance has been shaped by evolutions in financial models. Both the fee structure and the broader business model have also become ways to manage tensions between stakeholders. Previously, different stakeholders were billed using different fee structures, e.g., which caused friction. By providing a consistent and standardized method, the new model is seen as a way to manage both existing and new relationships, becoming a part of the governance framework.

> This is finally what I believe is the stable foundation for Dryad to approach growth with confidence because we're not approaching one institution and saying, yeah, I know they got a different deal, or going to the publishers and saying, I know I can't explain that. Now we have a much more stable foundation, a more consistent conversation to have, and it creates the ability now to reach into new stakeholders, new markets, new sectors. (P1)

5 Discussion

This analysis demonstrated that funding open infrastructure is never a simple process, but is one which requires reconfigurations of relationships, business models, and revenue streams. While "thinking like a business" can be a necessary strategy, it is one that affects other critical factors influencing sustainability (Section 4.2). While we have detailed the case of Dryad, sustainably funding ODIs is a system-wide problem that many infrastructures face (Imker, 2020; Skinner et al., 2025). We therefore conclude by discussing tensions which emerge when "thinking like a business" along four lines, which can be useful for open infrastructures more broadly.

5.1 Thinking like a business is not a bad thing, but it is not the only thing.

Within a (non-profit) organization, tensions can emerge when different logics and values come together, i.e. tensions between the need to procure funding and community values (Anheier, 2014; Ebrahim, 2019). Such tensions are visible within our analysis, for example in how Dryad needed to mobilize shared values to enroll institutions (Section 4.1.2). Our analysis suggests that careful implementation of business models can be a means to productively work with these tensions. Further, due to their flexible nature, business models can also act as boundary objects, where they can be adapted and take on new meaning through how activities are performed in local contexts (Doganova & Muniesa, 2015). The example of Dryad's business model can therefore usefully travel to other organizations and infrastructures.

Different forms of flexibility in Dryad's business model are also apparent in our analysis. Dyrad's partnership with CDL, which led to new technology and customers (Section 4.1.2), as well as the development of the new fee structure (Section 4.2) exemplified large changes in services and revenue models, or "Type A flexibilities" (Eschenfelder et al., 2022). Smaller adjustments (Type B flexibilities) also occurred, such as continued applications for new grants (Figure 3). Both these larger and more minute changes shaped core aspects important in sustaining an infrastructure, such as articulating values, defining community, and constructing governance structures (Section 4.2; Mounier and Dumas Primbault, 2023). This suggests that financial planning needs to carefully consider how business models and asset creation will affect other aspects of an open infrastructure, to decide what change is acceptable, and to recognize that sometimes large changes are needed to keep an infrastructure operational.

5.2 Thinking about relationships is part of thinking like a business.

ODIs are often encouraged to cultivate a mix of funding streams (Organisation for Economic Co-operation and Development (OECD), 2017); our analysis suggests that having a mix of relationships at various levels of maturity is also important. A combination of established, developing, and considered positioning relationships with customers, competitors and collaborators can help to embed an infrastructure in different webs of relations and practices (Section 4.1). Our analysis shows different ways of relating to various actors: emphasizing shared values with institutions; creating integrations for publishing workflows; or engaging in co-opetition with potential competitors. This suggests that a key to these different relationships is identifying what is of value to different actors and leveraging that in a mutually beneficial way.

Relationships with both publishers and the CDL took significant time to establish; other short-term relationships, within projects or collaborations, were also leveraged to meet specific goals, e.g. adopting shared metadata standards (Section 4.1.3). This suggests that relationships with different timescales are important in sustaining infrastructure, as are considerations about when relationships should end (Section 4.1.4). This mirrors broader discussions about when infrastructures themselves should be shut down (Cohn, 2016; POSI Adopters, 2025); especially when there are many similar infrastructures competing for the same funds (Section 4.1.3). While short-term infrastructures could be designed to support particular missions or other longer-term infrastructure (Knopp & Grenz, 2026), new relationships, in the form of mergers or consolidations could also be part of the solution (Skinner et al., 2025).

Relationships can be managed, and communities cultivated, through business models themselves (Section 4.2.2; Zott and Amit, 2010). For example, Dryad's new fee structure was seen to provide a means of mediating conversations; engaging stakeholders in the value proposition research also created both a sense of community and engendered support for the revenue model. Given the dynamic nature of both business models and infrastructure (Karasti & Blomberg, 2018; Zott & Amit, 2010), engaging around business models could be a continued form of dynamic community development.

5.3 Governing curation services as an asset

Treating curation services as an asset, something to generate durable income, changes how Dryad will relate to and govern this service (Birch, 2017, 2024). Curation services are part of the variable data publication charge, creating potential vulnerabilities in staffing dynamics. If not enough (or too many) datasets are received, it could necessitate firing or hiring staff. Assetizing curation services may also create a need to make curation more efficient to generate sufficient revenues, e.g. through implementing AI tools (Section 4.2.1; Lippincott et al., 2026). Focusing on turnaround (or implementing AI) could lead to having less capacity to provide human connection and support, which is seen to be both valuable and a value of Dryad (Section 4.2.1). Such questions will continue to be negotiated as the model is implemented, leading to potentially new processes of assetization, where perhaps the real asset will emerge to be human connection, rather than curation.

Governance structures also change as business models evolve and infrastructures scale (Jones et al., 2023; Treloar & Woodford, 2025). At Dryad, governance changed from community governance representative of researchers to a model based on stakeholder (customer) representation, composed mostly of publishing and institutional representatives (Section 4.2.2). Stakeholders are now an important part of Dryad's community and will make decisions about how the asset of curation is put into practice and governed. Maintaining the balance between perspectives of these different stakeholders, as new assets are created, will continue to be important to balance emerging tensions.

5.4 Valuing openness

While openness emerged as a differentiating factor for Dryad, it also complicated the development of the fee structure. Being an OI restricted some types of revenue models, while also enabling others, such as funding from SCOSS. Our findings demonstrate the persistent tension between the idea that open content and infrastructure should be freely available and the fact that openness requires investment (Section 4.2.1; Skinner et al., 2025). Valuing openness suggests that stakeholders need to realize that it takes time, money, and human labour to create useful open content (Leonelli, 2016).

The new fee structure follows many of the recommendations for ODIs, i.e. charging for services rather than for data access and shifting the burden of payment to data depositors (Organisation for Economic Co-operation and Development (OECD), 2017; POSI Adopters, 2025). This has some similarities to other forms of charging for open content, such as gold open access (OA) publication models, where commercial publishers charge authors (or their institutions) to make articles available (Butler et al., 2023). While gold OA increases access to content, it still raises concerns around affordability and equity (Valdez et al., 2024); against this backdrop, the growth of commercial data storage and publication platforms (Acker et al., 2026) could be a cause for concern. Publishing data might also be difficult for institutions to budget for, as data publication charges are typically dynamic, depending on systemic factors such as how much research an institution can conduct. This makes it challenging for institutions to follow normal budgeting workflows. More flexible processes of budgeting on the part of institutions for data publishing costs may need to be considered.

6 Conclusion

The case of Dryad provides an example of one set of practices for working towards financial sustainability that can be useful to other ODIs. This study has also contributed to opening discussions about how to sustainably fund open infrastructure and has made visible the relationships existing between various actors in doing so. This suggests that funding ODIs is not something that only one actor or infrastructure can solve, but is rather something that requires system-wide effort and consideration.

Footnotes

[1]Dryad API: https://datadryad.org/api; last accessed: 2026-03-04

[2]DataCite API: https://support.datacite.org/docs/api; last accessed: 2026-03-04

[3]DataCite - Consuming Citations and References: https://support.datacite.org/docs/consuming-citations-and-references; last accessed: 2026-03-04

[4]OpenAlex API documentation: https://docs.openalex.org/how-to-use-the-api/api-overview; last accessed: 2026-03-04

References

Acker, A., Struett, T., Tian, Y., & Finn, M. (2026). Platformization of research data infrastructure: A decadal analysis of scientific data management plans. *Big Data & Society*, *13*(2), 20539517261431593. https://doi.org/10.1177/20539517261431593

Amit, R., & Zott, C. (2001). Value creation in E-business. *Strategic Management Journal*, *22*(6-7), 493–520. https://doi.org/10.1002/smj.187

Anheier, H. K. (2014). *Nonprofit Organizations* (2nd Edition). Routledge. https://doi.org/10.4324/9781315851044

ATLAS.ti. (2022). https://atlasti.com

Attwood, T. K., Agit, B., & Ellis, L. B. M. (2015). Longevity of Biological Databases. *EMBnet.journal*, *21*, e803–e803. https://doi.org/10.14806/ej.21.0.803

Battilana, J., & Lee, M. (2014). Advancing Research on Hybrid Organizing – Insights from the Study of Social Enterprises. *The Academy of Management Annals*, *8*(1), 397–441. https://doi.org/10.1080/19416520.2014.893615

Birch, K. (2017). Rethinking Value in the Bio-economy: Finance, Assetization, and the Management of Value. *Science, Technology, & Human Values*, *42*(3), 460–490. https://doi.org/10.1177/0162243916661633

Birch, K. (2020). Technoscience Rent: Toward a Theory of Rentiership for Technoscientific Capitalism. *Science, Technology, & Human Values*, *45*(1), 3–33. https://doi.org/10.1177/0162243919829567

Birch, K. (2024). Assetization as a mode of techno-economic governance: Knowledge, education and personal data in the UN's System of National Accounts. *Economy and Society*, *53*(1), 15–38. https://doi.org/10.1080/03085147.2023.2264064

Bobrov, E., Broadhurst, R., Byers, N., Erdmann, C., Genova, F., Hellström, M., Kemp, J., Puebla, I., & FORCE11 Data Usage Typologies Working Group. (2026). Typology of Data Uses: An output of the FORCE11 Data Usage Typologies Working Group. https://doi.org/10.5281/zenodo.19184004

Borgman, C. L., Darch, P. T., Pasquetto, I. V., & Wofford, M. F. (2020). Our knowledge of knowledge infrastructures: Lessons learned and future directions. Retrieved December 14, 2023, from https://escholarship.org/uc/item/9rm6b7d4

Borgman, C. L., & Groth, P. (2025). From Data Creator to Data Reuser: Distance Matters. *Harvard Data Science Review*, *7*(2), 1–73. https://doi.org/10.1162/99608f92.35d32cfc

Borgman, C. L., Sands, A. E., Darch, P. T., & Golshan, M. S. (2016). The durability and fragility of knowledge infrastructures: Lessons learned from astronomy. *Proceedings of the Association for Information Science and Technology*, *53*(1), 1–10. https://doi.org/10.1002/pra2.2016.14505301057

Borgman, C. L., & Wofford, M. F. (2021). From Data Processes to Data Products: Knowledge Infrastructures in Astronomy. *Harvard Data Science Review*, *3*(3). https://doi.org/10.1162/99608f92.4e792052

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Braun, V., & Clarke, V. (2019). Reflecting on reflexive thematic analysis. *Qualitative Research in Sport, Exercise and Health*, *11*(4), 589–597. https://doi.org/10.1080/2159676X.2019.1628806

Butler, L.-A., Matthias, L., Simard, M.-A., Mongeon, P., & Haustein, S. (2023). The oligopoly's shift to open access: How the big five academic publishers profit from article processing charges. *Quantitative Science Studies*, *4*(4), 778–799. https://doi.org/10.1162/qss_a_00272

Callon, M., Méadel, C., & Rabeharisoa, V. (2002). The economy of qualities. *Economy and Society*, *31*(2), 194–217. https://doi.org/10.1080/03085140220123126

Cohen, A., Ivanovic, B., Iarkaeva, A., Nachev, V., & Bobrov, E. (2026). Matching data references and institutional output to map the reuse of biomedical research data [preprint]. https://doi.org/10.31222/osf.io/z9kjf_v2

Cohn, M. L. (2016). Convivial Decay: Entangled Lifetimes in a Geriatric Infrastructure. *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing*, 1511–1523. https://doi.org/10.1145/2818048.2820077

DataCite & Make Data Count. (2025). Data Citation Corpus Data File. https://doi.org/10.5281/zenodo.14897662

Doganova, L., & Muniesa, F. (2015). Capitalization Devices: Business Models and the Renewal of Markets. In *Making Things Valuable* (pp. 109–125). Oxford Academic. https://doi.org/10.1093/acprof:oso/9780198712282.003.0006

Dreyfuss, R., & Frankel, S. (2015). From Incentive to Commodity to Asset: How International Law is Reconceptualizing Intellectual Property. *Michigan Journal of International Law*, *36*(4), 557–602. https://repository.law.umich.edu/mjil/vol36/iss4/1

Dryad. (n.d.-a). Joint Data Archiving Policy (JDAP). Retrieved April 2, 2026, from https://datadryad.org/docs/JointDataArchivingPolicy.pdf

Dryad. (n.d.-b). Who we are. Retrieved April 2, 2026, from https://datadryad.org/about

Ebrahim, A. (2019). *Measuring Social Change*. Stanford University Press.

Edwards, P. N. (2010). *A vast machine: Computer models, climate data, and the politics of global warming*. MIT press.

Edwards, P. N., Jackson, S. J., Bowker, G. C., & Knobel, C. P. (2007). Understanding Infrastructure: Dynamics, Tensions, and Design. Retrieved August 11, 2023, from http://deepblue.lib.umich.edu/handle/2027.42/49353

Eschenfelder, K. R., Shankar, K., & Downey, G. (2022). The financial maintenance of social science data archives: Four case studies of long-term infrastructure work. *Journal of the Association for Information Science and Technology*, *73*(12), 1723–1740. https://doi.org/10.1002/asi.24691

Fenlon, K., Reza, A., Grimmer, J., & Wagner, T. (2023). Mutual Sustainability among Communities and Their Knowledge Infrastructures. *Proceedings of the Association for Information Science and Technology*, *60*(1), 133–144. https://doi.org/10.1002/pra2.775

Fjeldstad, Ø. D., & Snow, C. C. (2018). Business models and organization design. *Long Range Planning*, *51*(1), 32–39. https://doi.org/10.1016/j.lrp.2017.07.008

Frank, R. D. (2024). Constructing risk in trustworthy digital repositories. *Journal of Documentation*, *80*(6), 1508–1527. https://doi.org/10.1108/JD-08-2023-0157

Friedman, M. (1970). A Friedman doctrine– The Social Responsibility of Business Is to Increase Its Profits. *The New York Times*. Retrieved April 2, 2026, from https://www.nytimes.com/1970/09/13/archives/a-friedman-doctrine-the-social-responsibility-of-business-is-to.html

Gregory, K., Zurbach, J., Shankar, K., Mayernik, M., Campbell-Verduyn, M., Bezuidenhout, L., & Treloar, A. (2026). Sustaining Knowledge Infrastructures: Asking Questions and Listening for Answers. *International Journal of Digital Curation*, *20*(1). https://doi.org/10.2218/ijdc.v20i1.1069

Hansmann, H. (1980). The Role of Nonprofit Enterprise. *The Yale Law Journal*, *89*(5), 835–901. Retrieved April 2, 2026, from https://yalelawjournal.org/article/the-role-of-nonprofit-enterprise

Haupka, N., Culbert, J., Donner, P., Jahn, N., Lenke, C., Mayr, P., Meier, A., Mittermaier, B., Scheidt, B., Stahlschmidt, S., & Taubert, N. (2025). OPENBIB: Selected curated open metadata based on OpenAlex [dataset]. https://doi.org/10.5281/zenodo.15308680

Hooft, R. W. W., & Roos, M. (2025). Financing Models for Sustainable Data Reuse Infrastructure. *Data Science Journal*, *24*, 29. https://doi.org/10.5334/dsj-2025-029

Imker, H. J. (2020). Who Bears the Burden of Long-Lived Molecular Biology Databases? *Data Science Journal*, *19*(1), 8. https://doi.org/10.5334/dsj-2020-008

Imker, H. J. (2025). Sustainability During Instability: Long-Lived Life Science Databases and Science Funding Outlook in the United States [preprint]. https://doi.org/10.1101/2025.10.08.680785

Jones, S., Leggott, M., Lopez Albacete, J., Madalli, D., Pascu, C., Payne, K., Schouppe, M., & Treloar, A. (2023). Global Open Research Commons IG - RDA. https://doi.org/10.15497/RDA00087

Karasti, H., & Blomberg, J. (2018). Studying Infrastructuring Ethnographically. *Computer Supported Cooperative Work (CSCW)*, *27*(2), 233–265. https://doi.org/10.1007/s10606-017-9296-7

Kitchin, R., Collins, S., & Frost, D. (2015). Funding models for Open Access digital data repositories (P. G.E. Gorman And Professor Jennifer Rowley, Ed.). *Online Information Review*, *39*(5), 664–681. https://doi.org/10.1108/OIR-01-2015-0031

Knopp, P., & Grenz, T. (2026). Temporary Infrastructure: Temporal Boundary Work, Form Investments, and Delayed Endings. *Science, Technology, & Human Values*, 01622439261437666. https://doi.org/10.1177/01622439261437666

LaDonna, K. A., Artino, A. R., & Balmer, D. F. (2021). Beyond the Guise of Saturation: Rigor and Qualitative Interview Data. *Journal of Graduate Medical Education*, *13*(5), 607–611. https://doi.org/10.4300/JGME-D-21-00752.1

Leonelli, S. (2016). *Data-Centric Biology: A Philosophical Study*. University of Chicago Press.

Lippincott, S., Collister, L., & Skinner, K. (2026). Sustaining the Commons in the AI Economy: A Landscape Scan of Challenges and Strategies for Bridging AI Companies and Open Curated Collections. https://doi.org/10.5281/zenodo.19458128

Mayernik, M. S. (2023). Toward stronger coupling between technical infrastructures and institutional processes in data-intensive science. In X. Ma, M. Mookerjee, L. Hsu, & D. Hills (Eds.), *Recent Advancement in Geoinformatics and Data Science* (Vol. 558). Geological Society of America. https://doi.org/10.1130/2022.2558(04)

Mohr, A. H., Carlson, J., Ge, L., Herndon, J., Kozlowski, W., Moore, J., Petters, J., Taylor, S., & Vitale, C. H. (2024). Making Research Data Publicly Accessible: Estimates of Institutional & Researcher Expenses. https://doi.org/10.29242/report.radsexpense2024

Morselli, F., Touber, J., & Scharnhorst, A. (2025). Fostering Data Communities – perspective from a Data Archive Service Provider [preprint]. https://doi.org/10.48550/ARXIV.2502.02321

Mounier, P., & Dumas Primbault, S. (2023). Sustaining Knowledge and Governing its Infrastructure in the Digital Age: An Integrated View. https://doi.org/10.5281/zenodo.10036402

Muniesa, F., & Doganova, L. (2020). The time that money requires: Use of the future and critique of the present in financial valuation. *Finance and Society*, *6*(2), 95–113. https://doi.org/10.2218/finsoc.v6i2.5269

Okune, A. (2019). Decolonizing scholarly data and publishing infrastructures. Retrieved March 29, 2026, from https://blogs.lse.ac.uk/africaatlse/2019/05/29/decolonizing-scholarly-data-and-publishing-infrastructures/

Organisation for Economic Co-operation and Development (OECD). (2017). Business models for sustainable research data repositories. Retrieved May 21, 2025, from https://www.oecd.org/en/publications/business-models-for-sustainable-research-data-repositories_302b12bb-en.html

Porter, M. E. (1985). *Competitive Advantage: Creating and Sustaining Superior Performance*. Free Press.

POSI Adopters. (2025). The Principles of Open Scholarly Infrastructure v2.0. https://doi.org/10.14454/G8WV-VM65

Recker, J., Pegelow, L., Schuster, A., & Mauer, R. (2026). Hope for the best, plan for the worst: Reducing risks associated with repository cessation. *International Journal of Digital Curation*, *20*(1). https://doi.org/10.2218/ijdc.v20i1.1091

Ribes, D., Finholt, T., & University of Michigan. (2009). The Long Now of Technology Infrastructure: Articulating Tensions in Development. *Journal of the Association for Information Systems*, *10*(5), 375–398. https://doi.org/10.17705/1jais.00199

Riordan, D., Huang, C.-K., Neylon, C., & Steinhart, G. (2025). Data for: The State of Open Infrastructure Grant Funding, 2025 State of Open Infrastructure Report [dataset]. https://doi.org/10.5281/zenodo.15198421

Salamon, L. M. (1995). *Partners in Public Service: Government-Nonprofit Relations in the Modern Welfare State*. Johns Hopkins University Press. https://doi.org/10.56021/9780801849626

Shankar, K., & Eschenfelder, K. R. (2015). Sustaining Data Archives over Time: Lessons from the Organizational Studies Literature. *New Review of Information Networking*, *20*(1-2), 248–254. https://doi.org/10.1080/13614576.2015.1111699

Skelcher, C., & Smith, S. R. (2015). Theorizing Hybridity: Institutional Logics, Complex Organizations, and Actor Identities: The Case of Nonprofits. *Public Administration*, *93*(2), 433–448. https://doi.org/10.1111/padm.12105

Skinner, K., Collister, L., Huang, C.-K. (, Kemp, J., Lippincott, S., Neylon, C., Riordan, D., Sellanga, J., Steinhart, G., Thaney, K., Tsang, E., & Wu, C. (2025). 2025 State of Open Infrastructure: Trends in characteristics, funding, policy, and community health. https://doi.org/10.5281/zenodo.15198874

Star, S. L., & Griesemer, J. R. (1989). Institutional Ecology, 'Translations' and Boundary Objects: Amateurs and Professionals in Berkeley's Museum of Vertebrate Zoology, 1907-39. *Social Studies of Science*, *19*(3), 387–420. https://doi.org/10.1177/030631289019003001

Strecker, D., Pampel, H., Schabinger, R., & Weisweiler, N. L. (2023). Disappearing repositories: Taking an infrastructure perspective on the long-term availability of research data. *Quantitative Science Studies*, *4*(4), 839–856. https://doi.org/10.1162/qss_a_00277

Thaney, K. (2025). What We Talk About When We Talk About Open Infrastructure. https://doi.org/10.1146/katina-01152025-1

Thomer, A. K., & Rayburn, A. J. (2024). "A Patchwork of Data Systems": Quilting as an Analytic Lens and Stabilizing Practice for Knowledge Infrastructures. *Science, Technology, & Human Values*, *49*(5), 1073–1102. https://doi.org/10.1177/01622439231175535

Treloar, A., & Woodford, C. (2025). Global Open Research Commons: Enabling Curation for the Next 20 Years. *International Journal of Digital Curation*, *19*(1), 13. https://doi.org/10.2218/ijdc.v19i1.1054

Uffen, H., & Kinkel, T. (2019). Controlling the costs of long-term digital accessibility - A cost model for long-term digital accessibility. Retrieved April 2, 2026, from https://www.dpconline.org/blog/controlling-costs-of-ltdp

UNESCO. (2021). UNESCO Recommendation on Open Science. https://doi.org/10.54677/MNMH8546

U.S. National Library of Medicine. (n.d.). About NIH-Supported Data Sharing Resources. Retrieved April 2, 2026, from https://www.nlm.nih.gov/NIHbmic/bmic-about.html

Valdez, J. W., Vergara, L. C., Orihuela, G., & Fernandez, M. (2024). Overcoming the Tropical Andes publication divide: Insights from local researchers on challenges and solutions. *PLOS ONE*, *19*(6), e0306189. https://doi.org/10.1371/journal.pone.0306189

Vasileiou, K., Barnett, J., Thorpe, S., & Young, T. (2018). Characterising and justifying sample size sufficiency in interview-based studies: Systematic analysis of qualitative health research over a 15-year period. *BMC Medical Research Methodology*, *18*(1), 148. https://doi.org/10.1186/s12874-018-0594-7

Zott, C., & Amit, R. (2010). Business Model Design: An Activity System Perspective. *Long Range Planning*, *43*(2), 216–226. https://doi.org/10.1016/j.lrp.2009.07.004